Time Series Analysis of Air Pollutant Emissions and Electrical Access in Kenya

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UNITED STATES INTERNATIONAL UNIVERSITY - AFRICA

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A Research Project Submitted to School of Science and Technology in Partial Fulfillment of Requirements for the Award of Degree of Bachelor of Science in Data Science and Analytics

UNITED STATES INTERNATIONAL UNIVERSITY- AFRICA

SS 2026

STUDENT'S DECLARATION

I, the undersigned, declare that this research is my original work, except for cited quotations and references that has properly been acknowledged and referenced in accordance with the USIU-Africa requirements. Additionally, this research has not been submitted to any other college, institution, or any other university for examination, award of degree or publication or academic credit.

Signed..... ..... **Date**.....**05/12/2025**

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SUPERVISOR APPROVAL

This research proposal has been submitted for examination with my approval as the University Supervisor

Signed..... **Date**.....**05/12/2025**.....

Name of the Supervisor (ID Number)

Dr. Japheth Mursi

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ABSTRACT

This study aims to identify the correlation between demographic factors such as wealth and Kenya's ongoing electrification rates, alongside their impact on the environment, to inform as well as advise on future research or policy decisions. To measure the demographic, environmental, electrical access, and energy mix, it was decided that we would use GDP as the wealth demographic indicator, CO2 emissions as an environmental effects indicator, the population percentage that has access to electricity as the electrical access indicator, and fossil fuel energy contribution percentage combined with renewable energy generation output as our energy mix indicators.

The study identified the correlation between GDP and electrification rate to be greater than 0.9, the correlation between electrification and CO2 to be -0.55, and the correlation between renewable energy generation and CO2 emissions to be -0.51, through which we were able to reject the null hypothesis with a p-value less than 0.05, proving a correlation exists. While the time-series forecasting was conducted using generative transformers, which were used to generate predictive lines with the lowest MSE and highest R^2, the best model was a combination of CO2 and electrification, with values of 0.001923 and 0.452 respectively.

ACKNOWLEDGEMENT

I would like to express my gratitude to my supervisor, Dr. Japheth Mursi and Dr. Verrah A. Otiende for their feedback and guidance thought this research.

DEDICATION

To all those who seek to know, explore, and remain curious, even about the most trivial of revelations.

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LIST OF ACRONYMS

ADF – Augmented Dickey-Fuller

AIC – Akaike Information Criterion

CO_2 – Carbon Dioxide

EDA – Exploratory Data Analysis

EPRA – Energy and Petroleum Regulatory Authority

GDP – Gross Domestic Product

IEA – International Energy Agency

INDAAF – Instrumentation for the Detection of Atmospheric and Air Fluxes

IQR – Interquartile Range

IRENA – International Renewable Energy Agency

KPL – Kenya Power and Lighting Company

MSE – Mean Squared Error

NH_3 – Ammonia

NO_2 – Nitrogen Dioxide

OLS – Ordinary Least Squares

PCA – Principal Component Analysis

REA – Rural Electrification Authority

ROC – Receiver Operating Characteristic

R^2 – Coefficient of Determination

SDG7 – Sustainable Development Goal 7

SO_2 – Sulfur Dioxide

VIF – Variance Inflation Factor

WHO – World Health Organization

ARCH Test – Autoregressive Conditional Heteroskedasticity Test

Renewable Energy – Wind, Solar, Hydro, Geothermal

SDG – Sustainable Development Goals

ML – Machine Learning

CHAPTER 1

Introduction

1.1 Introduction and Background of the Study

A country's infrastructure for affordable electrical access is considered a hallmark of a prosperous and developing socio-economic environment. Kenya has also made progress in transitioning to greener energy solutions through initiatives like the Last Mile Connectivity Project and investments in the renewable energy sector (International Energy Agency [IEA], 2022; Kenya Power, 2020). Despite these changes, air pollution remains a concern, especially in urban areas like Mombasa and Nairobi, due to the increase in transport and industry. In rural areas, biomass fuels such as charcoal, kerosene, and firewood contribute to household air pollution (Gachanja et al., 2019).

In addition to this is the usage of fossil fuels during peak demand or outages (Ministry of Energy, 2021), leading to this study's time-series analysis to evaluate the relationship between electrification and air pollution trends in Kenya, aiming to generate insights for policymakers and stakeholders.

1.2 Problem Statement

Kenya has significantly increased electricity access from approximately 11% in the 1990s to over 76% by 2023 (World Bank, 1993–2023). At the same time, renewable energy accounts for about 90% of electricity generation (IEA, 2024). However, this transition does not necessarily guarantee reductions in pollutant emissions.

Many households still rely on biomass fuels, especially for cooking (Shilenje et al., 2022). Additionally, rising income levels may increase the use of energy-intensive appliances and fossil fuel-based technologies such as generators and vehicles.

These complexities highlight the need for empirical investigation into whether electrification correlates with reduced emissions in Kenya.

1.3 Objectives of the Study

General Objective

To investigate the correlation between electrification and changes in air pollutant emissions in Kenya.

Specific Objectives

1. To examine the composition of Kenya's electrical generation, particularly the fossil fuels vs renewable energy share composition.
2. To evaluate the influence of how demographic factors such as population and wealth affect energy use patterns, thereby the environmental impact of electrification.
3. To determine whether increased electrification correlates with reductions in ambient air pollution.
4. To provide insights on future trends that can be used to create relevant findings based on empirical findings.

1.4 Research Questions

1. What is the composition of Kenya's electrical generation, and how has the fossil fuels versus renewable energy source changed over time?
2. How do demographic factors such as population growth and wealth correlate with electrical usage access patterns and their environmental impacts?
3. How does the change in electrification correlate with reductions in ambient air pollution?
4. Based on historical trends, what are the possible future patterns of electrification and how can they inform public policies?

1.5 Research Justification

This research aims to challenge or affirm the assumption that increasing electrification reduces environmental impacts caused by biomass fuel use. It also examines how technological advancement and rising affluence influence energy use and emissions. The findings will support policymakers, researchers, and stakeholders in designing informed energy and environmental strategies.

1.6 Scope of the Study

The study was limited to within Kenyan borders, with the goal of conducting a nationwide analysis of ambient air pollution, and is not divided into regions, counties, or districts. Therefore, the data would include the national GDP per capita covering the time period from the 1990s to 2023. It would also include other energy mix data such as fossil fuel percentage generation and renewable energy generation from the time period 1990s to 2023.

The research aims to study ambient air pollution within Kenyan borders; therefore, the following parameters were selected to be used as indicators of ambient air pollution levels: CO2 as the main indicator. Such gases, usually produced through the burning of fossil fuels, are either health hazards or environmental pollutants in excess.

The dataset would require a time-series analysis; therefore, data ranging from historical to recent periods is required. The time scope of the study would include 1990 to 2025, and data outside these ranges would not be considered for use in the study.

The tools used to prepare, process, and build data models will include Python and Excel. Excel would be used to handle simple data cleaning operations, while Python provides the framework to further clean, build, train, test, and evaluate models.

1.7 Limitations

There are several limitations that were encountered when conducting the research, such as data availability. Data held by local companies or generally considered historically irrelevant are hard to find publicly and are often kept by key institutions. Furthermore, when data was sourced, it was discovered that they do not always match when intercompared, such as discrepancies in time and regional coverage.

1.8 Chapter Summary

This chapter introduced the study, outlining its purpose, objectives, and scope. It establishes the foundation for analysing electrification and pollution trends in Kenya.

CHAPTER 2

Literature Review

2.1 Introduction

This chapter introduces existing literature on energy generation and composition, electrification, demographic influences on electrical usage, and its environmental impacts on pollution. The review combines multiple locally published reports from the energy, environmental, and governing sectors over the past 5 years to provide a coherent study that establishes the empirical and theoretical framework that interlinks the correlation between Kenya's electrification progress and national pollutant emissions. The chapter concludes by identifying research gaps that motivate the time-series analysis this research hopes to undertake.

2.2 Theoretical Review

There are three major theories that serve as the backbone and source of contest for this study, through which our study hopes to investigate their impacts.

The Energy Ladder Model (Hoiser & Dowd, 1987; Barnes & Floor, 1996), which proposes that as income rises, households tend to move to newer and cleaner energy sources, while the Energy Stacking Theory (Masera, Saatkamp & Kammen, 2000), an alternative to the Energy Ladder Model, suggests that as a household's income increases, they may not abandon old energy sources (traditional energy sources), but instead stack the newer ones on top of them.

However, the Sustainable Energy Transition Theory (Geels, 2002) proposes that the transition from high-carbon fuels to low-carbon fuels is influenced by a variety of factors, including

infrastructure, social behaviors, regulations, etc. Therefore, it cannot be judged solely by one variable.

It is with the above considerations that the study aims to address the gap created by these theories in a way that is applicable to the Kenyan environment. It hopes to discover the correlation between income and renewable energy generation and how that translates to Kenya's electrification status in consideration of energy generation mix ratios. It further aims to address whether forecasts can be made with or without the influence of external factors.

2.3 Empirical Review

Reports from the International Energy Agency (IEA, 2022; 2024) and the Energy and Petroleum Regulatory Authority (EPRA) indicate that while Kenya's energy sector is predominantly occupied by renewable energy sources, fossil fuels are still used as backups for peak hours and outages. Shilenje et al. (2022) found that while electrification replaces kerosene usage for lighting and warming, it has limited impacts on cooking practices, which has led to household air pollution persisting.

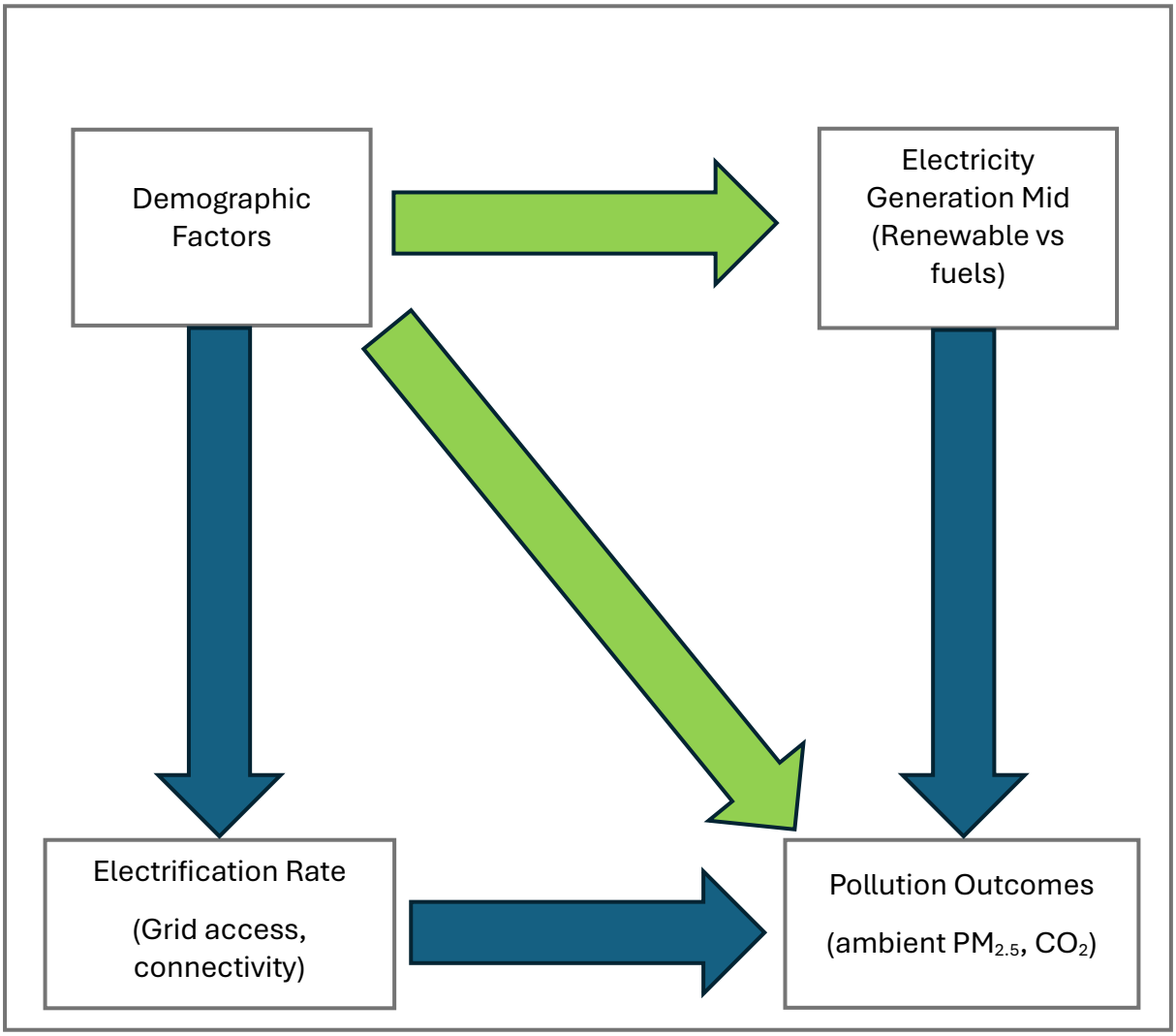
These reports indicate the challenge of a household's access to the electrical grid in lowering pollution rates and the persistent cultural behaviors that may diminish or invalidate the benefits that electrification may bring. Thereby, creating a gap in our understanding of whether the grid generates more pollution than it solves or whether electrification in households actually reduces pollution levels.

Further reports by the National Environment Management Authority (NEMA, 2020) indicate that air pollution levels are higher in urban and industrial areas, possibly due to the increase in industry and transportation emissions. This report demonstrates a gap and showcases how affluence and wealth affect CO2 emissions.

2.4 Conceptual Review

Understanding the interactions between the study's variables, which are electrification, demographics (GDP), energy mix generation, and air pollution, would be crucial to the modelling phase. Figure 1 shows this interaction visually by representing the direct relationships suspected or discovered in blue, while indirect relationships are shown in green as theorized based on the covered literature.

Figure 1: Conceptual Diagram



2.4.1 Electrification Trends and Energy Access in Kenya

The rise in electrification from 15% in 1990 to over 70% by 2020 (World Bank, 2021) and the expansion of solar systems (Acker & Kammen, 2020) have played a major role in increasing electrification access across multiple households.

However, despite these gains and improvements, there still exists a disparity between urban and rural areas of the country, where some counties lag behind (Ondiek, 2019). Therefore, research objective two requires the study to examine how the uneven distribution of wealth affects electrification rate patterns.

2.4.2 Composition of Electrical Generation

Clean energy sources make up approximately 80 percent of the total electrical output (IEA, 2022; Mburu et al., 2019), with geothermal providing the backbone of all baseload power. While fossil fuels are only a minor fraction of the energy output, they are primarily used as backup for peak load management and off-grid stabilization (Kenya Power & Lighting Company, 2022).

This renewable energy mix places Kenya among the leading countries in clean energy in Africa. However, the reliance on fossil fuels to maintain electrical stability may hinder the transition and continue to contribute to rising pollutant emissions. Therefore, examining the energy mix composition over time may help bridge this knowledge gap in understanding the environmental impacts of electrification at a nationwide scale and how this interacts with the energy generation mix, as stated in research objective one.

2.4.3 Demographic and Socioeconomic Factors in Determination of Energy Use

It has been observed that urban household consume two to three times more electricity than rural households due their lifestyle habits and cost of living, indicating the need to purchase more electrical appliance to support their lifestyle and living standard (Nyangena & Stage, 2020). In conjunction with this, it was observed that with a raising population and income levels, there was an increased aggregate electrical demand (Peters & Sievert, 2016)

This once again underscores the importance of research objective 2, highlighting how demographic factors like wealth influence and population growth can affect energy use and pollution outcomes.

2.4.4 Electrification and Ambient Air Pollution

The complex relationship between electrification and air pollution makes it difficult to research how one affects the other. Household air pollution declines when clean energy replaces traditional energy, which remains limited in Kenya due to behavioural habits (Ezzati & Kammen, 2001). Ambient air pollution, on the other hand, is more influenced by transport and industry, with locations that have industries and high traffic having higher pollution rates than those without (Kiai, Kanali, Sang, & Gatari, 2021).

These studies highlight a key knowledge gap pertaining to the scarcity of empirical studies in Kenya that integrate electrification and long-term pollution trends. Addressing this gap can reveal whether increased electrical access correlates with a reduction in ambient air pollution.

2.4.5 Projection and Forecasting in Advising Policy

Projections indicate that Kenya will continue to expand its renewable electrical energy generation capacity, particularly in the geothermal and wind sectors, over the next few decades (IEA & IRENA, 2022). However, population growth, urbanization, and increasing vehicle ownership are similarly expected to increase, thereby increasing electrical demands and transport emissions simultaneously (NEMA, 2020). If electricity demand increases faster than renewable supplies, reliance on thermal generators may be necessary, thereby increasing pollution emissions.

In response to the above speculation, long-term planning that integrates demographic, energy, and pollution data is essential to ensure sustainable energy and environmental outcomes. This is in line with research objective 4 and research question 4, which mandate the need to provide empirical evidence based on historical analysis that can forecast future trend patterns and inform policymakers about environmental and energy policy decisions.

2.5 Chapter Summary

This chapter covers some of the key knowledge gaps that this research aims to address, most of which are centered around the following points: electrification does not always lead to a decrease in air pollution, as some literature shows that people may simply switch to other activities that still burn the same amount of energy; the lack of studies that focus on Kenya's unique energy transition phase, electrification drive, and demographic circumstances; and the lack of long-term trend analysis of these correlations, as well as predictive models.

CHAPTER 3

Methodology

3.1 Introduction

This chapter outlines the design philosophy and the steps the research has taken to arrive at the end results. It describes the research designs, data collections methods, analysis techniques and other design considerations such ethical consideration. Through this chapter, the reader is expected to gain an understanding of design process and validation methods the research employed to generate and verify the results extracted from the models.

3.2 Design Methodology

The research follows a typical Data Science Methodology structure to guide the model development and analysis process. The methodology is designed to align with the research objectives, which focus on analysing the relationship between CO2 emissions, electrification rates, energy generation mix, and GDP in Kenya, as well as using the acquired data to predict future trends.

The analytical workflow consists of the following key stages:

Problem Understanding: The study aims to understand the correlation between CO2 emissions, electrification rates, energy generation, and GDP, while also predicting future trends.

Analytical Approach: A combination of quantitative and predictive analytical approaches was chosen due to the need to analyse relationships using quantitative methods, while predictive models are applied for time series forecasting.

Data Requirements: The study utilises data spanning from 1990 to 2025 for both predictive and analytical purposes. The required datasets include CO2 emissions, electrification rates, energy generation mix, and GDP data, all of which have to be contextually relevant to Kenya.

Data Preparation: Similar collected datasets were merged, while unrelated data were dropped from the datasets, using Excel and Python respectively. To handle missing data, forward interpolation techniques were employed in the Python environment.

Data Modelling: Several models were implemented, including Pearson correlation analysis to discover possible relationships, while other techniques, such as using generative transformer models, are used for time series tasks.

Evaluation: The model's performance is evaluated using statistical metrics such as MSE, R^2, and p-value hypothesis testing, with an alpha significance of 0.05.

Deployment and Interpretation: The models are deployed to gather insights into the relationships between variables, with diagrammatic representation where possible, and to forecast future patterns in emissions and electrification trends.

3.3 Research Design

The research adopts quantitative, experimental, and time series approaches to determine the relationship between CO2 emissions, electrification rates, energy generation mix, and economic growth (GDP) in Kenya.

A quantitative approach was selected due to the study's dependence on large amounts of numerical data and the need to identify patterns and establish relationships between the variables. It does this by employing correlation techniques to assess both the strength and

direction of the relationships between CO2 emissions and electrification rates, electrification rates and CO2 emissions, and GDP and electrification rates.

In addition, an experimental approach was used to evaluate the feasibility of using regression models as alternative means of correlation assessment and time-series analysis. This was done because regression models are generally easier to understand and immediately deploy.

The research also adopts newer time-series analytical approaches such as generative transformer models, in addition to classical statistical approaches like Gamma-based modelling.

The combination of these techniques allows for both analysis of correlations between the variables of interest and the use of time series approaches to model the future trends of electrification and CO2 emissions.

3.4 Data Collection Methods

Data collection was done by web scraping through various publicly available reports and statistics found online, through which a clear rationale of how they obtained their data was given, or through institutionally recognized bodies like the World Bank. As the data is mainly focused locally in Kenya as compared to abroad (where more research institutes and organizations research and publish related data online for ease of public access) the data required for this research was obtained as an aggregated result of multiple sources.

World Bank: Provided electrification rates, population, GDP per capita, fossil-fuel generation, and renewable-energy-generation rates.

INDAAF: Provided monthly concentrations of air pollutants: sulphur dioxide, ammonia, nitric acid, nitrogen dioxide, and CO_2.

Electricity Maps: Provided daily data on CO_2 emissions for the past 4 years.

EPRA: Provided the peak energy demands of each year between 2019 and 2024 through the annual report they publish and make publicly available.

All the above resources are available online and can be accessed in their respective order through the links provided below:

World Bank:

https://data.worldbank.org/indicator/EG.ELC.RNWX.KH?end=2023&locations=KE&start=1993&view=chart

INDAAF:

https://indaaf.obs-mip.fr/catalogue/?checkBoxCriteria=%7B%22jsonAttributes%22%3A%5B%7B%22key%22%3A%22sites.shortName%22%2C%22values%22%3A%5B%22Mbita%2C+Kenya%22%5D%7D%5D%7D&currentSelection=9aa2158d-9691-4f68-8a35-acbe19f082e9

Electricity Maps (requires user registration):

https://app.electricitymaps.com/datasets?zone=KE

EPRA:

https://www.epra.go.ke/statistics-0

3.6 Data Analysis Methods

The study combines statistical and predictive data analysis methods to determine relationship between variables and forecast future trends.

Descriptive analysis was used to summarise data and extract key insights to its composition measuring mean, standard deviation and trends over time. This primarily helped inform the initial understanding of the data and possible applications.

This was followed by correlation and regression analysis, in which Pearson correlation coefficients were calculated and applied to determine the relationships between the variables of interest, followed by regression analysis to evaluate any linear dependent and independent relationship shared among the variables.

For predictive purposes, time-series forecasting techniques were implemented, with transformer models being employed to predict future trend chain patterns.

3.7 Validation and Evaluation

Transformer models are validated by dividing the data into two groups: the training data and the validation data. This is then employed to test how well the model performs on unseen data, preventing overfitting and ensuring the model's ability to perform as trained on unseen data.

The validation of the model's accuracy was tested using statistical techniques such as calculating their MSE and R^2 values. Which measure the prediction accuracy and model fit. Lower MSE indicate better model performance, while higher R^2 values indicate a stronger explanatory power of the model to explain the variability observed in the data. This technique was employed in both quantitative models and predictive models

To evaluate the significance of any correlation, and regression analytical techniques hypothesis testing methods were employed, specifically the use of the p-value and critical value threshold of 0.05 (alpha value). Relationships with p-values less than 0.05 are deemed statistically significant.

3.8 Ethical Considerations

The data was sourced from online datasets; therefore, no direct human involvement was included in the data collection process. As such, beyond appropriate referencing, no further ethical considerations are required on that front. Ethical consideration should be given to how the research results will be used as an informative aid and not as the sole basis for final or critical policy decisions.

3.9 Chapter Summary

This chapter details the general outline of how the research was conducted and how the results were generated and evaluated. It explains the rationale behind choosing different analytical approaches and evaluation methods and equips the reader with general knowledge of how the model was developed for reproducibility and educational purposes. It acts as a log of the steps taken and how the results have been validated and evaluated.

CHAPTER 4

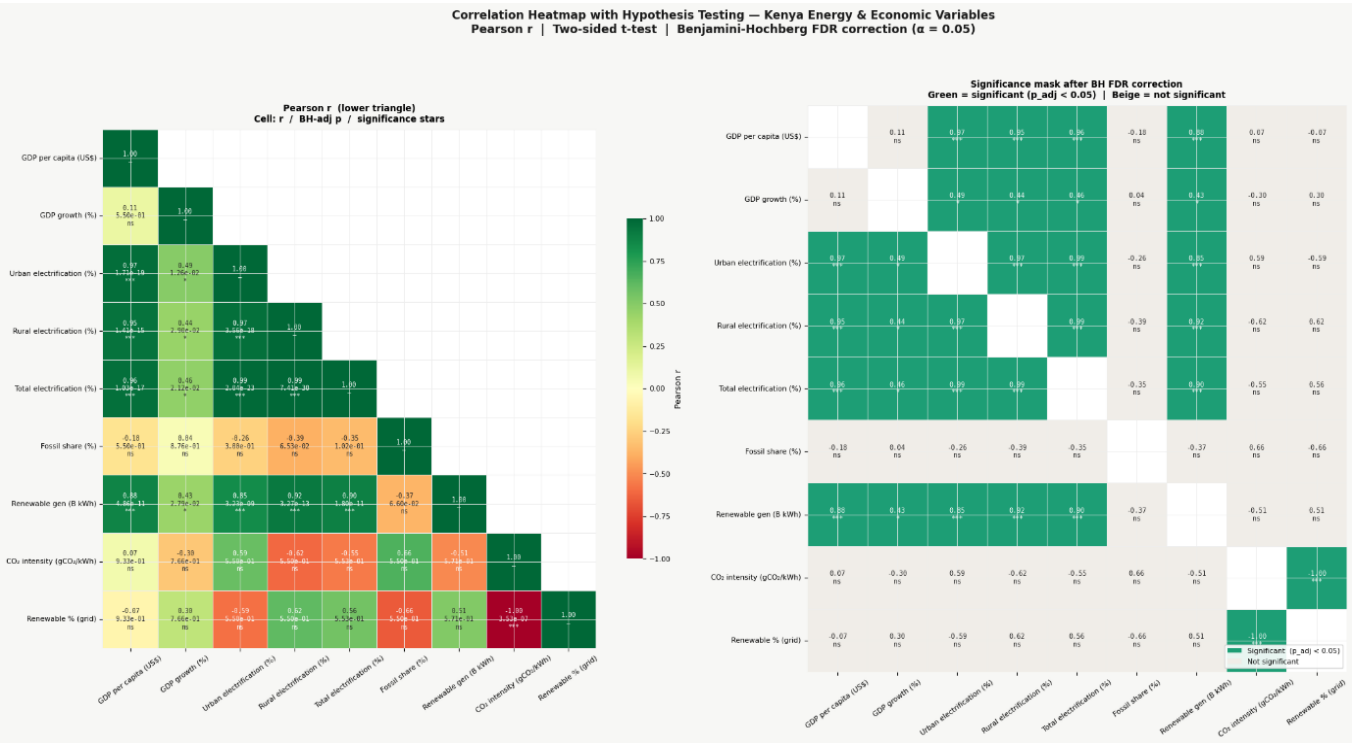
Design Results

4.1 Introduction

This chapter presents the results obtained from the investigation of the relationships between CO2 emissions, electrification rates, energy mix, and GDP in Kenya. The results are derived from statistical analysis, correlation findings, and predictive model outputs. It includes a brief analysis and discussion of the presented results and their relation to the research questions posed in Chapter 1.

4.2 Presentation of Results

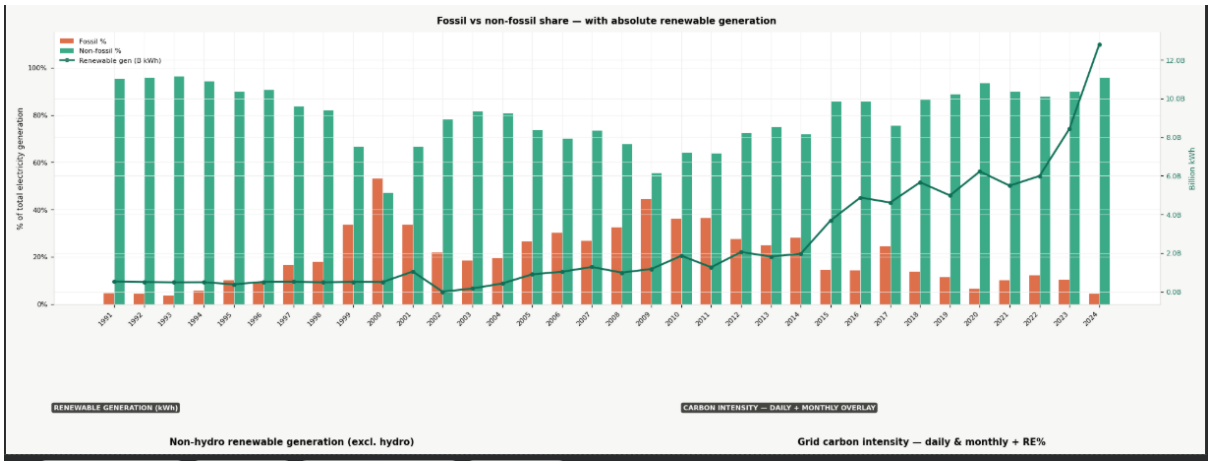
Figure 2: Pearson Correlation Heatmap and Hypothesis Test



The heatmap on the right shows the correlation between all variables, with stronger positive correlations appearing in green and stronger negative correlations appearing in red. In Figure

2 on the right, there is a representation of whether the hypothesis test confirms that the correlation is statistically significant, with green indicating the rejection of the null hypothesis, thereby confirming the correlation's significance.

Figure 3: Energy Generation Mix Change Overtime



In Figure 3, the bar graph shows the change in total electricity generation percentage (% of total energy generated). The green bars represent renewable energy sources, while the brown bars represent fossil fuel sources. The green line represents the total energy generation (kWh) output provided by renewable energy sources.

Regression Results

In this section, we present the regression results from different combinations of variables that either had a high correlation value or were directly necessary to answer the research questions investigating the relationships between the variables. To validate whether the use of regression was warranted, an OLS assumption test was conducted immediately afterward.

Figure 4: Regression GDP ~ Total Electrification

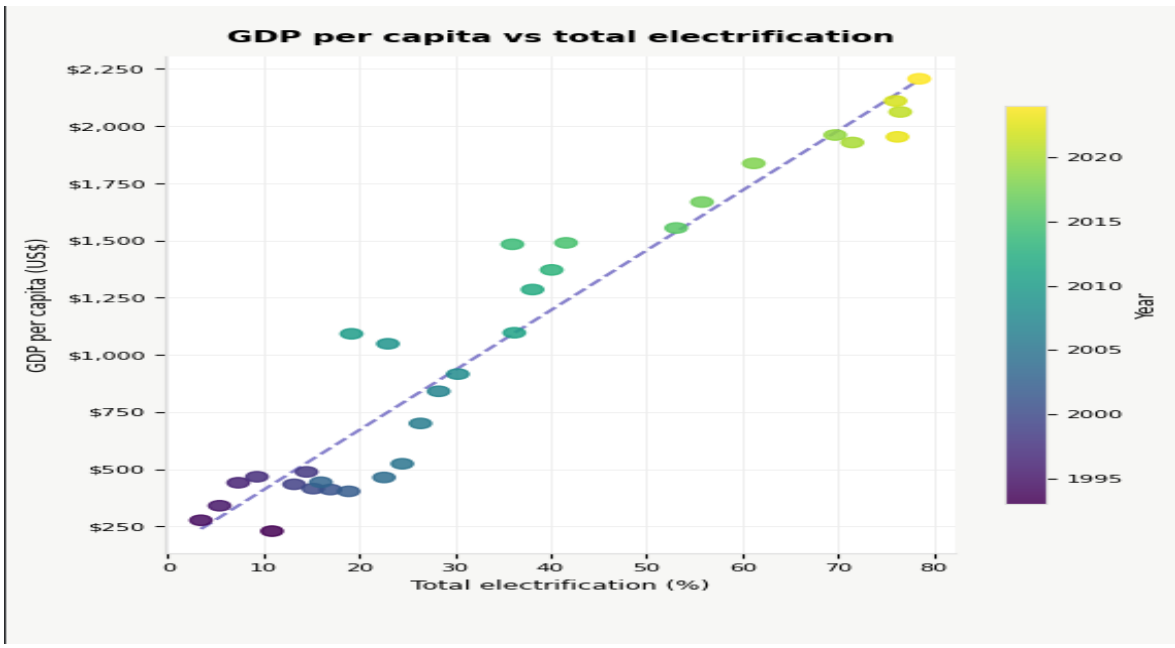


Figure 5: GDP ~ Total Electrification OLS Assumption Test

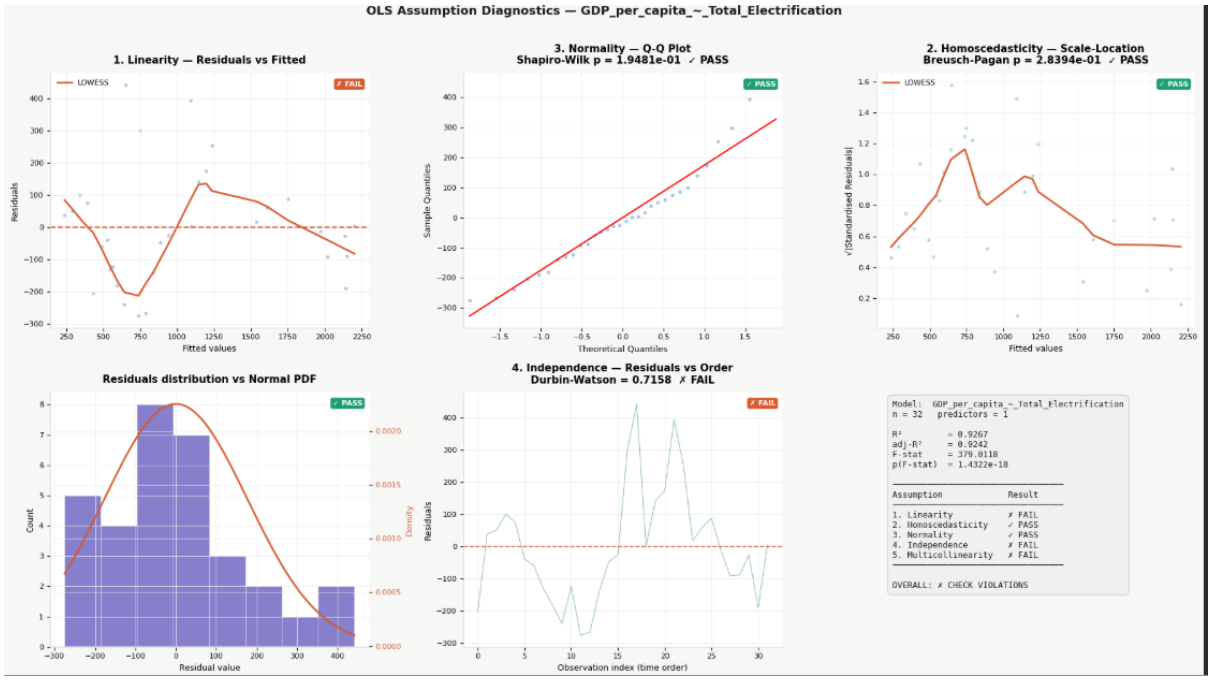


Figure 6: Renewable Gen vs CO_2 Intensity

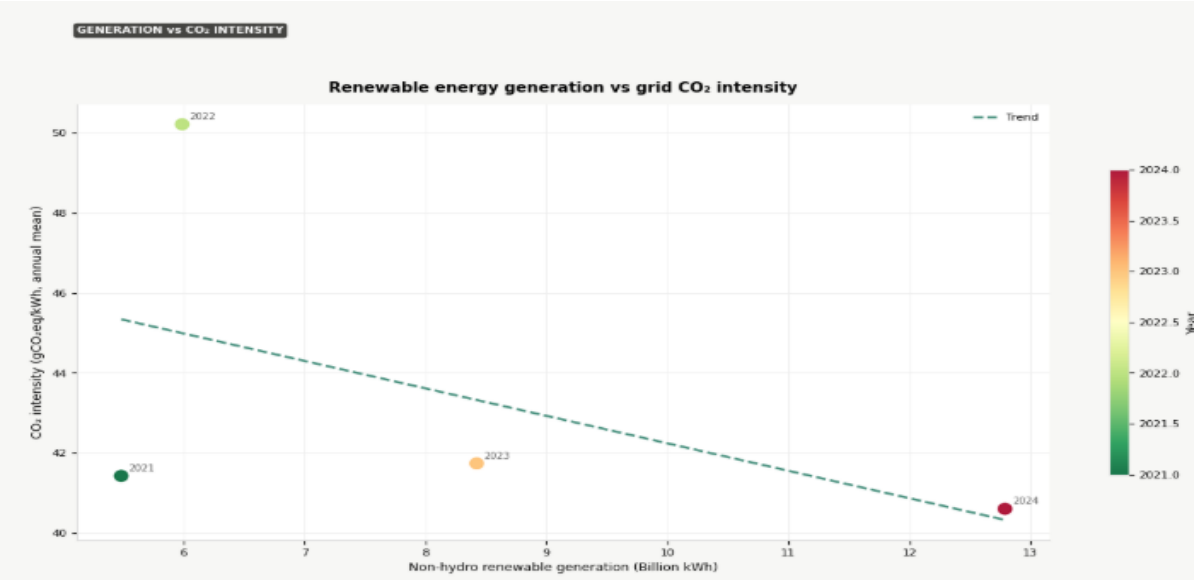


Figure 7:OLS Assumption Test for Renewable Gen vs CO2 Intensity

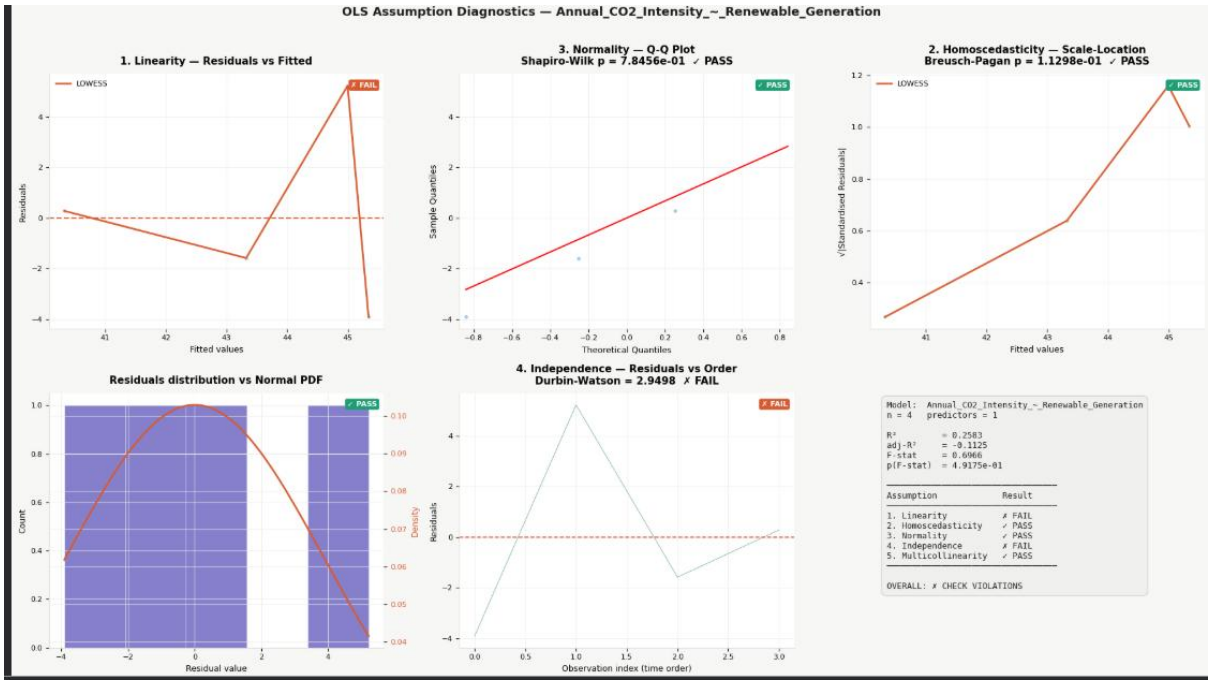


Figure 8: Renewable Gen (%) ~ CO_2 Emissions

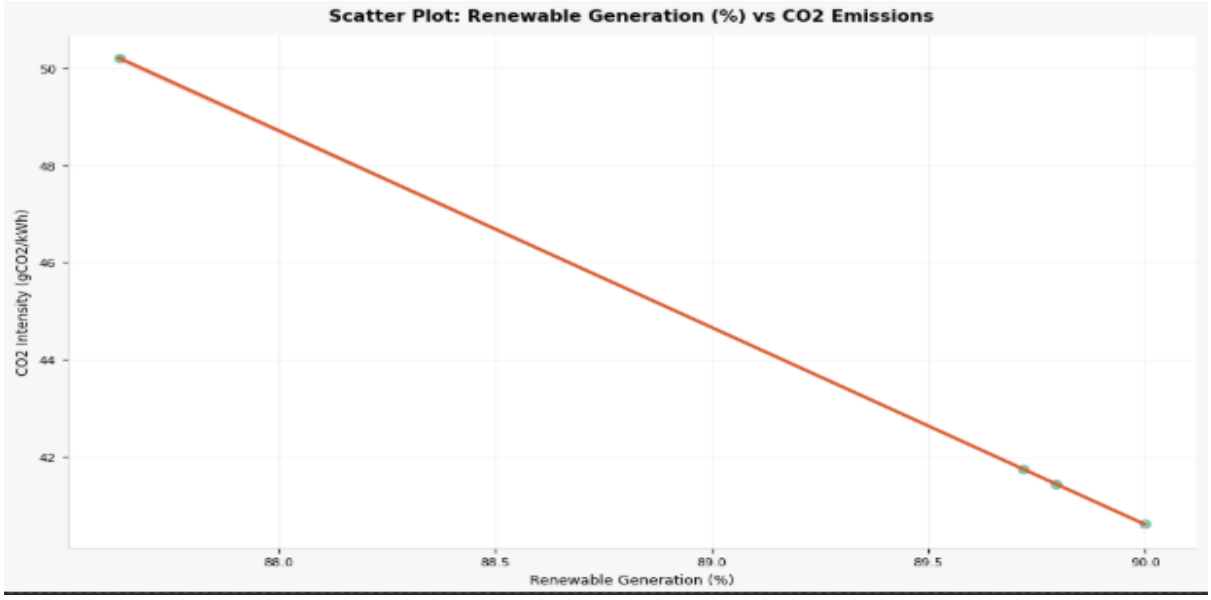


Figure 9: OLS Assumption Test for Renewable Gen (%) ~ CO_2 Emissions

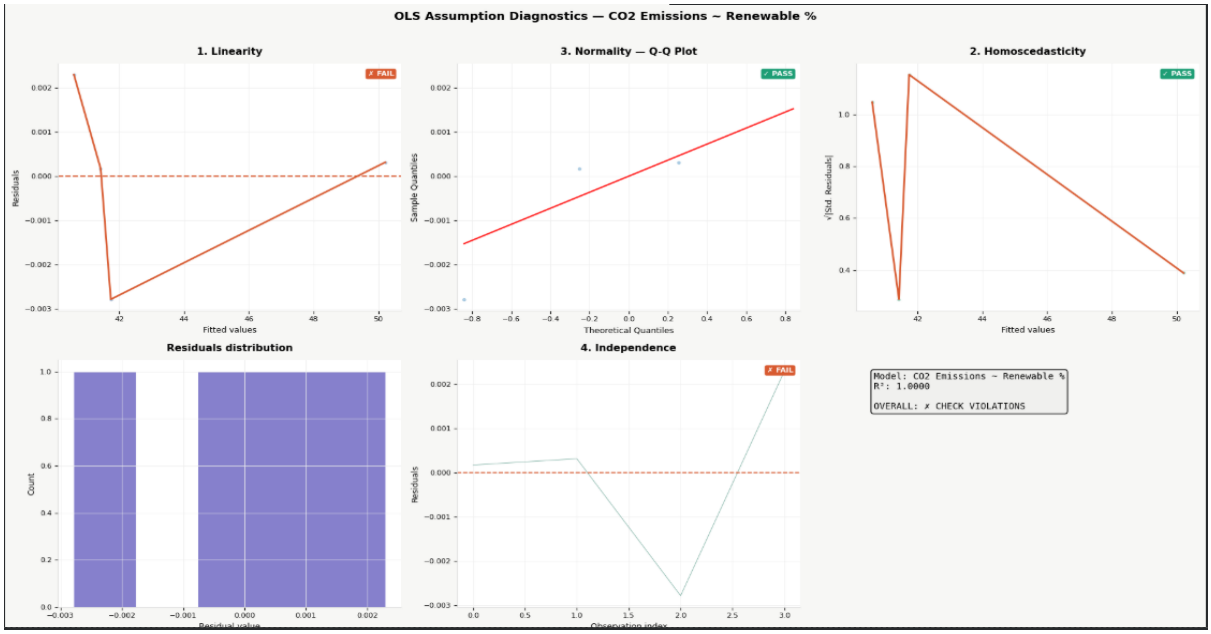


Figure 10: Total Electrification (%) ~ CO_2 Emissions

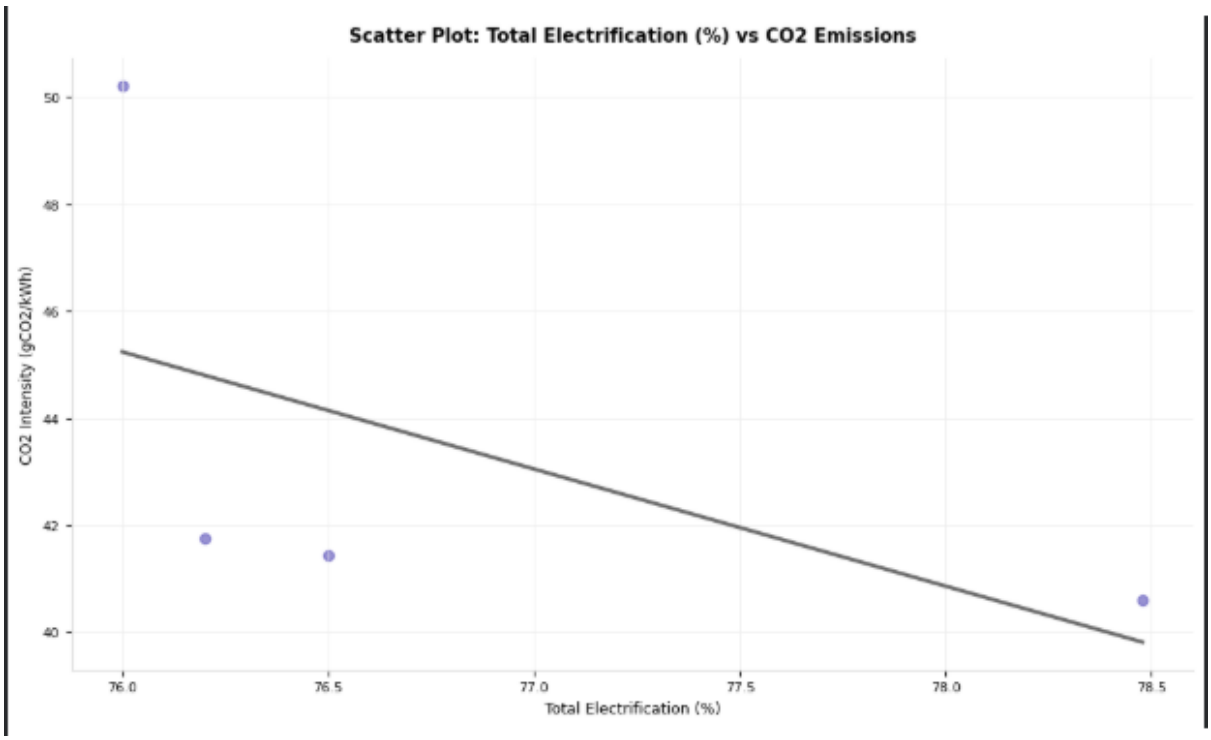
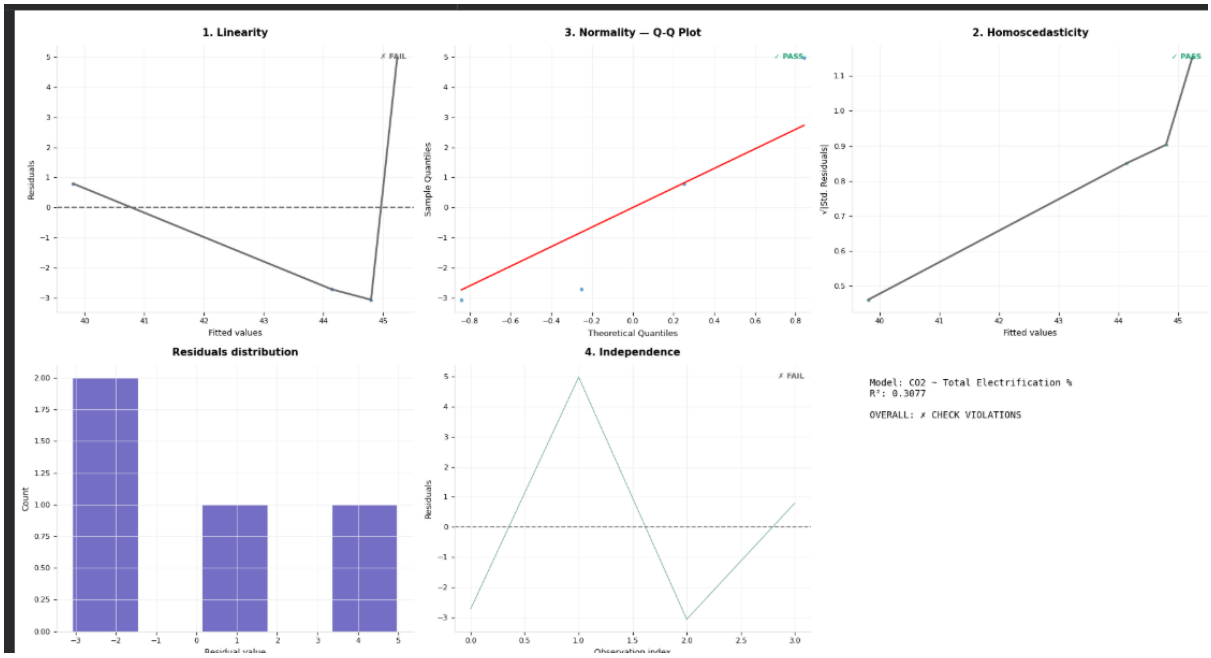


Figure 11: OLS Assumption Test for Total Electrification (%) ~ CO2 Emissions



Transformer Prediction Results.

The graphs in this section represent the outputs of the models when given the predictor variable and its position in time, and tasked with forecasting the expected change in the variable over the next 30 steps in the sequence. Figures 12 and 13 use daily CO2 as the predictor variable, with Figure 13 including electrification as an additional independent variable that may influence changes in CO2. The multiple lines (validation lines) in Figures 12 and 13 are due to the abundance of daily CO2 data, which allows for a larger validation dataset and, consequently, multiple segments to be included in the 30-step validation test.

Figure 12: Transformer Prediction (Daily CO_2)

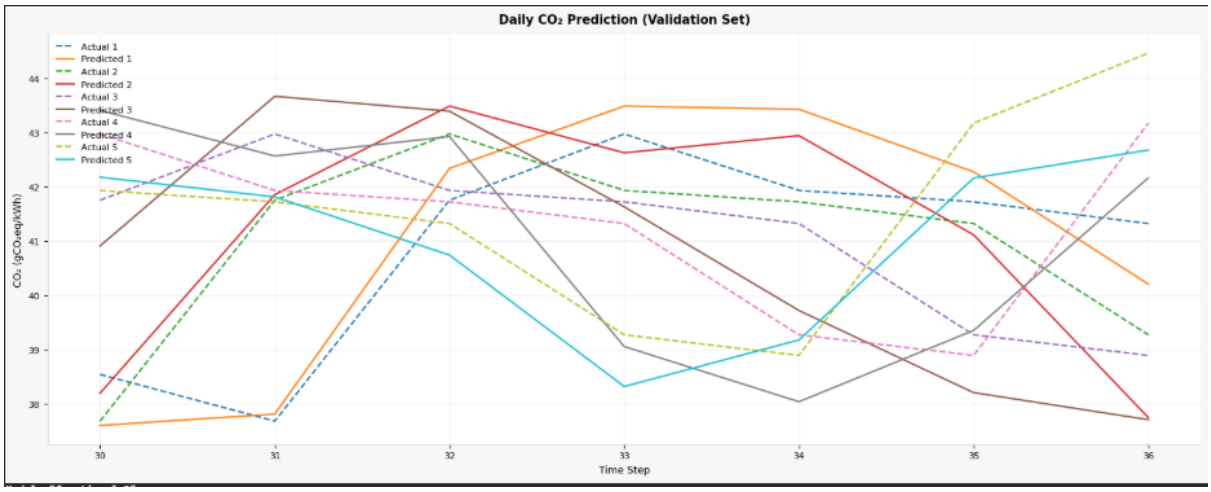


Figure 13: Transformer Prediction (Electrification + CO_2)

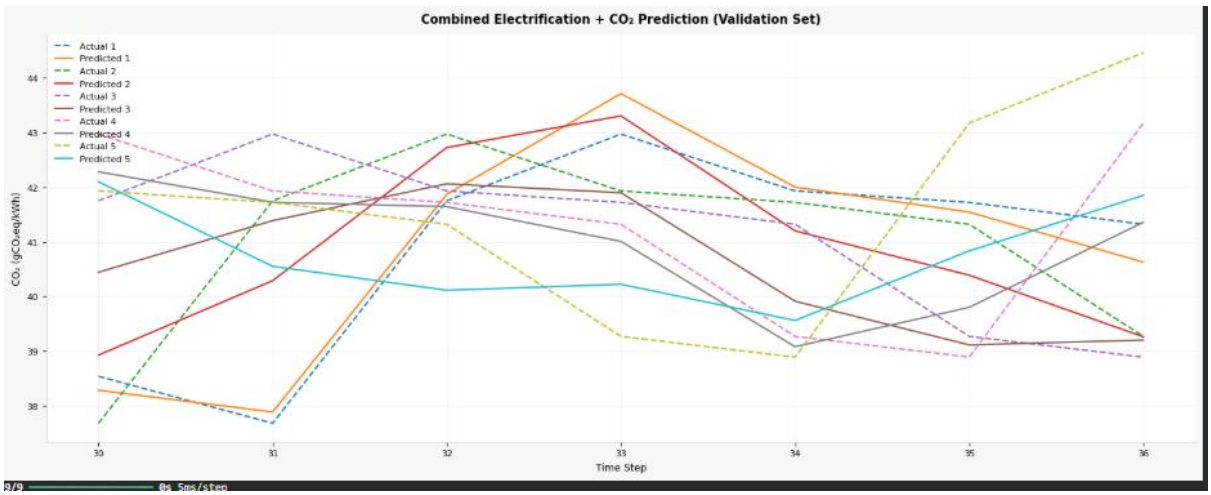


Table 1: Transformer Model Performance

TRANSFORMER MODEL PERFORMANCE		
Model	**Val MSE (scaled)**	**R^2 (original scale)**
Daily CO_2	0.001272	0.581229
Combined CO_2 + Elc	0.002804	0.477562

4.3 Analysis of Results

Figure 2 shows the correlations between all the variables. From it, we observe that GDP per capita has a strong positive correlation with electrification rate and renewable energy generation, indicating that an increase in economic wealth is strongly associated with increases in electrification rate and renewable energy generation.

Figure 2 also shows that both urban and rural electrification have a negative correlation with CO2 intensity, indicating that there may be an association such that as urban and rural electrification increases, CO2 intensity decreases accordingly.

From Figure 2, we also observe that CO2 intensity has a negative correlation with renewable energy generation and its percentage share, indicating that there may be an association such that as renewable energy generation and its share increase, CO2 intensity decreases.

CO2 emissions are very strongly negatively correlated with the percentage of renewable energy used for electricity generation. As seen in Figure 2, the correlation coefficient between these two variables is -1.

Figure 3 shows that the amount of renewable energy being generated is increasing, while the percentage share of total energy generated by renewable sources also shows signs of increasing, aside from two spikes in fossil fuel generation in 2000 and 2009.

Figures 4 to 11 show that OLS regression is not suitable for these variables, regardless of their correlation, due to failing to meet the basic assumptions required for valid OLS application. However, the results do agree with the correlation table shown above.

Figures 12 and 13 demonstrate the models' ability to predict future data based on past data that is generally similar to the actual data. This is shown in Table 1, as both models have low MSE, but the model responsible for Figure 12 performed better in terms of R^2 results.

4.4 Discussion of Findings

The findings from the analysis of the energy generation mix composition show that Kenya is mainly dependent on renewable energy sources to generate electricity, with fossil fuels serving as a supplementary source. We observed that, over time, the increase in power generated by renewable energy sources has continued to grow, while the fossil fuel contribution percentage to the energy grid has continued to decrease, despite the two spikes seen in the data. This indicates that, although the infrastructure has shifted towards renewable energy sources, fossil fuels may continue to serve as a backup in case of energy shortfalls or special circumstances. Furthermore, it also indicates that while Kenya has shifted from mainly using fossil fuels, it has yet to become completely independent of them.

In relation to demographic factors, the results show a strong positive correlation between GDP per capita and electrification rate, in addition to renewable energy generation. The results support the claim that cleaner energy growth may be directly tied to a country's affluence, which fits the Energy Ladder Model theory, where a country's wealth leads it to transition towards cleaner energy resources. However, the persistent fossil fuel usage observed supports the Energy Stacking Theory, meaning that an increase in income would not necessarily lead to the complete eradication of fossil fuel use as an energy source.

The negative correlation between CO2 emissions and electrification, as well as between CO2 emissions and renewable energy generation, suggests that increased electrification supported by renewable energy generation is associated with a decrease in CO2 emissions. However, the findings also indicate that this relationship needs to be supported by behavioural shifts from companies and households that may prefer using fossil fuels over renewable alternatives, as clean energy generation alone does not fully account for reductions in CO2 emissions. Instead, the percentage of people using it over fossil fuel alternatives shows a much stronger correlation.

The predictive analysis further contributes to understanding and modelling future trends. The model's ability to forecast CO2 emissions with relatively low MSE scores, as shown in Table 1, indicates the feasibility of this approach. It was observed that the model using only time-series-related data outperformed the electrification–CO2 hybrid model, which may indicate that time plays a larger role in changes in CO2 emissions compared to electrification, or that the imbalanced dataset between daily CO2 and electrical data impacted the hybrid model more negatively than expected.

4.5 Chapter Summary

From the above findings, it is clear that Kenya has made significant progress towards its goal of providing clean energy to its citizens, and that the impact on air pollution is influenced by a multitude of factors such as renewable energy generation percentage, electrification, and socio-economic variables.

CHAPTER 5

Evaluation, Validation and Deployment

5.1 Introduction

This chapter covers the validation and evaluation criteria and the process of developing the model, as well as the measures taken to ensure that the resulting insights are credible. It outlines the validation techniques used and the results obtained, and further discusses the implications of these results and the potential for model deployment.

5.2 Evaluation Criteria

The performance of the model and validation of data are assessed using both statistical and predictive accuracy metrics. For correlation analysis, statistical significance is determined using p-values, with a critical threshold of 0.05; for a result to be significant, the p-value is expected to be less than 0.05.

For both regression and predictive models, the R^2 value is used as a metric to determine how well the model explains the observed data variability, with lower values indicating poorer performance. In predictive models, both R^2 and MSE are used in tandem, with MSE measuring the average prediction error; lower values indicate a more suitable model, while higher values indicate poorer performance.

In addition, visual graphs are generated, as seen in Figures 12 and 13, which allow for visual verification of the model's consistency and how closely the predicted values align with the actual values.

5.3 Validation Process

The validation process involves applying both predictive and quantitative evaluation methods throughout the study and assessing the suitability of the models for deployment before any insights can be derived or used.

The regression models had their suitability tested using a checklist of OLS assumptions. These include tests for linearity, normality of residuals, homoscedasticity, and independence of errors.

For predictive models (the transformer), the dataset was split into training and testing subsets while preserving temporal order to ensure realistic sequential forecasting. The models were trained on historical data and tasked with reproducing forward, unseen segments of the series. This tests how well the model generalises to unseen data in order to predict future trends.

In addition, multiple validation sequences are used in the transformer models, as displayed in Figures 12 and 13, due to the size of the dataset, allowing for a more robust validation process.

5.4 Results of Validation

The regression models were found to have failed some of the OLS assumption tests, making any insights gained from them invalid for academic use. However, the results shown by the regression graphs are consistent with the correlation results.

The transformer models demonstrated strong predictive performance, as shown in Table 1, with the CO_2-only model showing a lower MSE (0.001272) and a higher R^2 value (0.581229), indicating better performance than the electrification–CO_2 combination model.

The combined model, however, showed slightly lower performance, with an MSE of 0.002804 and an R^2 of 0.477562.

Overall, these results show that, in this case, regression models are less suitable for handling time-series-related data than transformer models, which demonstrate a strong capability to capture general patterns.

5.5 Discussion of Validation

The regression models' failure to identify correlations between variables, especially their inability to pass the independence assumption, suggests that more complex variables are involved. This reinforces the need for more advanced modelling techniques when dealing with volatile and complex real-world data.

The transformer performed well and demonstrated its effectiveness in capturing 58% of the variability observed in the data. Upon visual inspection in Figures 12 and 13, the predicted lines mostly follow the general direction of the actual data.

These findings highlight the practicability of using correlation analysis to derive insights not only in static variables but also in time-dependent and more complex relationships, while predictive modelling required more complex approaches to achieve actionable results. The validation process therefore supports the use of transformer models for forecasting and predictive tasks.

5.6 Deployment

The models can be deployed to provide insights for policymakers and researchers on how expanding infrastructure, especially electrification, impacts the environment and its relationship with a country's wealth and energy composition.

To facilitate broader use, it would be best if the model were integrated into a data-driven dashboard capable of displaying future projections of emissions, electrification rates, and energy generation composition. Such a system can support policymakers and researchers in assessing the environmental impact of new policies and sustainable strategies. In addition, the models could be designed to update over time and retrain on newly available data.

5.7 Chapter Summary

This chapter presented the evaluation, validation, and deployment criteria and strategies that the project aimed to study and implement. Metrics such as R^2, MSE, and p-values (with an alpha threshold) were used to assess the suitability of the models. The validation results showed that the regression models were unsuitable for predictive tasks, while transformer models proved more suitable. Finally, the chapter outlined how the model can be deployed to aid policymakers and researchers in their decision-making, and what insights can be derived from this area of study.

CHAPTER 6

Discussion, Conclusions and Recommendations

6.1 Introduction

This chapter discusses the overall conclusions that can be drawn from this research and the recommendations of the study. It summarises the key findings of the research in relation to its objectives, highlights key limitations, and provides recommendations for further improvements based on key findings.

6.2 Research Summary

This study was conducted to investigate the relationship between electrification and air pollution in Kenya, and how this may be used to derive future environmental impacts. The research focused on analysing the energy generation composition and the influence of demographic factors such as population growth on CO2 emissions.

The approach utilised available data spanning from 1990 to 2025 for time-series-related tasks. Statistical methods such as correlation and regression analysis were used to examine relationships between variables, and transformers were used for predictive analysis tasks.

6.3 Results Interpretation and Discussion

The findings of the study reveal that Kenya's energy generation is mainly composed of renewable energy sources, but fossil fuels are still being used and may serve as supplementary sources during emergencies or peak periods. This indicates that Kenya is gradually transitioning towards a greener energy future with cleaner energy integrated into the national power grid.

From the results, a strong positive relationship was also observed between economic factors, renewable energy, and electrification, suggesting that increases in electrification may be dependent on a country's wealth and its ability to invest in infrastructure development. This is also reflected in the development of green energy generation solutions, as greater wealth may have enabled faster adoption. However, the continued use of fossil fuels alongside renewable energy highlights the Energy Stacking Theory, where households and industries utilise a mix of both renewable and fossil-fuel-based energy despite increasing affluence.

Furthermore, the study also identified a negative correlation between electrification and CO2 emissions, indicating that increased electrical access can contribute to lowering the carbon footprint. However, this relationship also depends on other factors such as the energy generation mix, suggesting that electrification alone is not sufficient to reduce pollution levels.

The predictive modelling demonstrated that transformers were well suited for forecasting future CO2 emission trends, although the results indicate that historical emission patterns play a more dominant role in determining future trends.

6.4 Limitations

The study encountered a few limitations that deserve acknowledgement.

First, the study relied heavily on secondary data sources, which resulted in gaps in the data, particularly in the early and later years.

Second, the models were limited by data availability, which may explain why some analyses produced more conclusive and clearer patterns than others.

Third, the study focused on Kenya in general and may have overlooked regional conditions and variations that could differ from, or represent outliers to, the national trend.

Finally, both the analytical and predictive models lacked specific data on household contributions to pollution levels, as well as more granular local and global CO2 emissions data.

6.5 Conclusions

This study concludes that electrification in Kenya is positively associated with reduced environmental impact and economic growth, but the data suggests that additional factors must be considered, as its effect on ambient air pollution is not straightforward.

While increased electrification and renewable energy usage contribute to lower CO_2 intensity, the continued reliance on fossil fuels and industrial activities limits the overall environmental benefits.

The study also concludes that traditional regression approaches are insufficient for modelling complex environmental relationships, whereas advanced models such as transformers provide more reliable predictive capabilities.

Overall, achieving significant reductions in air pollution requires a comprehensive approach that integrates electrification with clean energy adoption, industrial regulation, and behavioural change.

6.6 Recommendations

Based on the findings of this study, the following recommendations are proposed:

The promotion of behavioural change and awareness to ensure a reduction in household-level pollution rates can be achieved through incentives, campaigns, or general education.

Strengthen green energy expansion and explore alternatives such as batteries, potential dams, or similar systems to store energy instead of relying on fossil fuel backups.

Promote continued expansion of green energy technologies while implementing policies that encourage industries to switch to grid power or further limit emissions.

Increase electrification rates while simultaneously increasing the renewable energy portion of the power output.

6.7 Recommendations for Further Research

Future research should focus on more granular and detailed data to better understand the relationship between electrification and emissions.

In particular, researchers are encouraged to investigate average daily household CO_2 emissions through longitudinal (long-term) studies. Such studies would provide deeper insights into how household energy consumption patterns evolve over time and how they contribute to overall emissions.

Additionally, future studies could incorporate regional-level analysis to capture spatial differences in electrification and pollution. Expanding the range of variables to include transportation, industrial activity, and policy interventions would also improve the robustness of future models.

6.8 Chapter Summary

This chapter provided a comprehensive summary and discussion of the study's findings, highlighting the relationship between electrification, energy use, and air pollution in Kenya. The study concluded that while electrification contributes to cleaner energy use, its impact on emissions is influenced by multiple factors. Limitations were identified, and recommendations were proposed to support policy development and future research. These findings contribute to a deeper understanding of sustainable energy transitions in developing countries.

AI USE DISCLOSURE

AI was used for coding assistance and write-up refinement to make points clearer and ensure language accuracy and consistency throughout the document, with Claude and Manus agents used in code assistance and ChatGPT used in text refinement and literature lookup. All ideas, steps and logical conclusions, including the analysis, recommendations, and interpretations, are the authors' own unless otherwise cited.

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Appendices

Appendix 1: Dataset Samples

1.1: Daily CO_2 emissions

Table 2: Daily CO2 emissions

Datetime (UTC)	Country	Zone name	Zone id	Carbon int	Carbon int	Carbon-fre	Renewable	Data source	Data estim	Data estimation method
01/01/2021 00:00	Kenya	Kenya	KE	43.61	100.12	89.26	89.26	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
02/01/2021 00:00	Kenya	Kenya	KE	42.05	97.87	89.64	89.64	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
03/01/2021 00:00	Kenya	Kenya	KE	40.57	95.62	90.01	90.01	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
04/01/2021 00:00	Kenya	Kenya	KE	43.81	100.13	89.21	89.21	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
05/01/2021 00:00	Kenya	Kenya	KE	42.93	98.25	89.43	89.43	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
06/01/2021 00:00	Kenya	Kenya	KE	42.58	97.73	89.51	89.51	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
07/01/2021 00:00	Kenya	Kenya	KE	42.25	97.3	89.59	89.59	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
08/01/2021 00:00	Kenya	Kenya	KE	42.54	98.22	89.52	89.52	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
09/01/2021 00:00	Kenya	Kenya	KE	41.14	96.28	89.87	89.87	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
10/01/2021 00:00	Kenya	Kenya	KE	40.26	94.97	90.08	90.08	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
11/01/2021 00:00	Kenya	Kenya	KE	43.18	99.09	89.36	89.36	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
12/01/2021 00:00	Kenya	Kenya	KE	43.4	99.15	89.31	89.31	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
13/01/2021 00:00	Kenya	Kenya	KE	42.69	97.57	89.49	89.49	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
14/01/2021 00:00	Kenya	Kenya	KE	43.16	98.63	89.37	89.37	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
15/01/2021 00:00	Kenya	Kenya	KE	43.38	99.4	89.32	89.32	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
16/01/2021 00:00	Kenya	Kenya	KE	41.64	96.96	89.74	89.74	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
17/01/2021 00:00	Kenya	Kenya	KE	40.94	96.16	89.92	89.92	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
18/01/2021 00:00	Kenya	Kenya	KE	44.17	100.76	89.12	89.12	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
19/01/2021 00:00	Kenya	Kenya	KE	43.25	98.95	89.35	89.35	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
20/01/2021 00:00	Kenya	Kenya	KE	42.92	98.32	89.43	89.43	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
21/01/2021 00:00	Kenya	Kenya	KE	40.95	94.41	89.92	89.92	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
22/01/2021 00:00	Kenya	Kenya	KE	41.46	95.53	89.79	89.79	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
23/01/2021 00:00	Kenya	Kenya	KE	39.38	92.55	90.3	90.3	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
24/01/2021 00:00	Kenya	Kenya	KE	38.06	90.99	90.62	90.62	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
25/01/2021 00:00	Kenya	Kenya	KE	41.49	95.62	89.78	89.78	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
26/01/2021 00:00	Kenya	Kenya	KE	43.61	99.57	89.26	89.26	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL
27/01/2021 00:00	Kenya	Kenya	KE	43.43	99.22	89.3	89.3	IEA + Electricity Maps Estimation	TRUE	ESTIMATED_GENERAL_PURPOSE_ZONE_MODEL

1.2: Annual GDP per Capita

Table 3: Annual GDP per Capita

Country Name	Country Code	Indicator Name	Indicator Code	1960	1961	1962	1963	1964	1965	1966	1967	1968	1969	1970	1971	1972
Aruba	ABW	GDP per capita growth	NY.GDP.PCAP.KD.ZG													
Africa Eastern and S	AFE	GDP per capita growth	NY.GDP.PCAP.KD.ZG		-2.13294	5.008777	2.794386	1.829575	2.245805	1.917422	2.418486	1.185892	2.098682	-1.74436	2.477833	-0.10909
Afghanistan	AFG	GDP per capita growth	NY.GDP.PCAP.KD.ZG													
Africa Western and	AFW	GDP per capita growth	NY.GDP.PCAP.KD.ZG		-0.25114	1.538396	4.741446	3.08762	1.82372	-3.70351	-11.0058	-0.75191	12.46265	14.6106	7.815939	0.655533
Angola	AGO	GDP per capita growth	NY.GDP.PCAP.KD.ZG													
Albania	ALB	GDP per capita growth	NY.GDP.PCAP.KD.ZG													
Andorra	AND	GDP per capita growth	NY.GDP.PCAP.KD.ZG												-2.50059	1.012624
Arab World	ARB	GDP per capita growth	NY.GDP.PCAP.KD.ZG								3.801321	7.643216	4.387701	19.45527	4.470432	10.74876
United Arab Emirate	ARE	GDP per capita growth	NY.GDP.PCAP.KD.ZG												16.83943	21.0227
Argentina	ARG	GDP per capita growth	NY.GDP.PCAP.KD.ZG		3.697198	-2.48121	-6.85164	8.371062	8.853456	-2.16954	1.629723	3.230851	8.008685	1.450585	4.007823	0.032009
Armenia	ARM	GDP per capita growth	NY.GDP.PCAP.KD.ZG													
American Samoa	ASM	GDP per capita growth	NY.GDP.PCAP.KD.ZG													
Antigua and Barbud	ATG	GDP per capita growth	NY.GDP.PCAP.KD.ZG													
Australia	AUS	GDP per capita growth	NY.GDP.PCAP.KD.ZG		0.463351	-1.14808	4.198557	4.9013	3.923448	0.067987	4.97145	3.255945	4.828076	5.084796	0.543808	2.017874
Austria	AUT	GDP per capita growth	NY.GDP.PCAP.KD.ZG		4.960717	2.02147	3.471466	5.419336	2.810013	4.904479	2.24101	3.931242	5.909496	6.749435	4.646943	5.592381
Azerbaijan	AZE	GDP per capita growth	NY.GDP.PCAP.KD.ZG													
Burundi	BDI	GDP per capita growth	NY.GDP.PCAP.KD.ZG		-15.902	6.331875	1.964997	3.28867	1.130321	2.123023	10.96864	-2.64849	-3.57614	18.71931	0.27971	-6.3998
Belgium	BEL	GDP per capita growth	NY.GDP.PCAP.KD.ZG		4.630257	4.794035	3.574354	5.949139	2.624445	2.461462	3.291576	3.785044	6.328284	6.247846	3.796489	4.884088
Benin	BEN	GDP per capita growth	NY.GDP.PCAP.KD.ZG		1.451144	-5.11105	2.825281	4.63548	3.234782	1.500862	-0.98609	1.674366	0.682677	-0.12853	-3.68733	4.011658
Burkina Faso	BFA	GDP per capita growth	NY.GDP.PCAP.KD.ZG		2.550233	4.587499	-2.73015	0.721438	2.117328	-1.09581	7.016897	1.315165	0.276667	-1.61623	-0.3224	0.563182
Bangladesh	BGD	GDP per capita growth	NY.GDP.PCAP.KD.ZG		3.11042	2.434774	-3.31559	7.760333	-1.31823	-0.37247	-4.72039	6.313411	-1.61576	2.920797	-7.11748	-15.6125
Bulgaria	BGR	GDP per capita growth	NY.GDP.PCAP.KD.ZG													
Bahrain	BHR	GDP per capita growth	NY.GDP.PCAP.KD.ZG												-1.00159	5.851353

1.3 Annual Electrical Access (%)

Table 4: Annual Electrical Access (%)

Country Name	Country Cc	Indicator Name	Indicator Code	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002
Aruba	ABW	Access to electricity (% of population)	EG.ELC.ACCS.ZS	100	99.15366	99.19713	99.23991	100	100	100	100	100	100	91.7	100	100
Africa Eastern and Southern	AFE	Access to electricity (% of population)	EG.ELC.ACCS.ZS											19.91726	19.94428	21.54865
Afghanistan	AFG	Access to electricity (% of population)	EG.ELC.ACCS.ZS											4.4	9.3	14.1
Africa Western and Central	AFW	Access to electricity (% of population)	EG.ELC.ACCS.ZS				31.65347	32.69221	33.69484	32.69739	32.52392	33.5135	34.89386	34.24612	35.32327	35.80911
Angola	AGO	Access to electricity (% of population)	EG.ELC.ACCS.ZS											24.2	20	26.3
Albania	ALB	Access to electricity (% of population)	EG.ELC.ACCS.ZS	100	100	100	100	100	100	100	100	100	100	99.4	99.4	99.4
Andorra	AND	Access to electricity (% of population)	EG.ELC.ACCS.ZS	100	100	100	100	100	100	100	100	100	100	100	100	100
Arab World	ARB	Access to electricity (% of population)	EG.ELC.ACCS.ZS								76.61178	77.54126	78.1564	80.45867	81.35479	81.32362
United Arab Emirates	ARE	Access to electricity (% of population)	EG.ELC.ACCS.ZS	100	100	100	100	100	100	100	100	100	100	100	100	100
Argentina	ARG	Access to electricity (% of population)	EG.ELC.ACCS.ZS	92.1548	92.49219	92.82945	93.16605	93.50127	93.83447	94.16494	94.49201	94.815	95.13316	95.7	95.5	96.1
Armenia	ARM	Access to electricity (% of population)	EG.ELC.ACCS.ZS											98.9	100	98
American Samoa	ASM	Access to electricity (% of population)	EG.ELC.ACCS.ZS													
Antigua and Barbuda	ATG	Access to electricity (% of population)	EG.ELC.ACCS.ZS	95.76305	95.94065	96.11813	96.29494	96.47039	96.6438	96.81448	96.98177	97.14498	97.30335	97.7	97.7	100
Australia	AUS	Access to electricity (% of population)	EG.ELC.ACCS.ZS	100	100	100	100	100	100	100	100	100	100	100	100	100
Austria	AUT	Access to electricity (% of population)	EG.ELC.ACCS.ZS	100	100	100	100	100	100	100	100	100	100	100	100	100
Azerbaijan	AZE	Access to electricity (% of population)	EG.ELC.ACCS.ZS										97	98.9	98.6	100
Burundi	BDI	Access to electricity (% of population)	EG.ELC.ACCS.ZS									3.9	2.070949	2.5	2.8	3.1
Belgium	BEL	Access to electricity (% of population)	EG.ELC.ACCS.ZS	100	100	100	100	100	100	100	100	100	100	100	100	100
Benin	BEN	Access to electricity (% of population)	EG.ELC.ACCS.ZS							14.5	18.58963	19.57825	20.56204	21.5	21.9	23.4
Burkina Faso	BFA	Access to electricity (% of population)	EG.ELC.ACCS.ZS				6.1	6.12501	6.625333	7.122936	7.61714	8.107265	6.9	9.1	9.5	9.9
Bangladesh	BGD	Access to electricity (% of population)	EG.ELC.ACCS.ZS		14.29	9.905419	12.71788	17.8	18.33805	21.14439	22.4	26.74619	29.54021	32	35	37.8
Bulgaria	BGR	Access to electricity (% of population)	EG.ELC.ACCS.ZS	100	100	100	100	100	100	100	100	100	100	88.1	88.6	89.1
Bahrain	BHR	Access to electricity (% of population)	EG.ELC.ACCS.ZS	99.9414	99.96739	99.98718	99.99714	99.99977	100	100	100	100	100	100	100	100

Appendix 3: Budget

Item	Cost
Personnel	0
Hardware	0
Software	0
Transport	0
Datasets	0
Documentation	0
Total	Minimal

Appendix 4: Code Book

The link below contains the code book, data repository and files accessed throughout this study, including a readme file that explains the usage and introductory information:

https://github.com/CMK-Coll/DSA3900-Data-Repository

Appendix 5: IRB Ethical Review

The research was review and approved by ISERC on the 10th April 2026 with the approval application number being USIU-A/ISERC/US1416-2026. This approval certifies that the study has cleared all ethical considerations and dated to last till 10th April 2027. The signed approval letter can be found in the following GitHub link:

https://github.com/CMK-Coll/DSA3900-Data-Repository

Appendix 6: NACOSTI license

NACOSTI license was submitted and is currently pending approval.